

Econ 409 Final Report

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1. Introduction.

The asset our team will forecast is Microsoft (MSFT). In addition to the price series, we choose the Price Earnings Ratio (P/E Ratio) and Short Interest Ratio (SI Ratio) as our two fundamentals, which allows us to forecast the price of the Microsoft stock based on both valuation and market sentiment.

Given our chosen fundamentals, we aim to develop a trading model that generates buy or sell signals based on deviations from historical averages. When the P/E ratio indicates undervaluation and the SI ratio suggests elevated bearish sentiment, we consider it a buy signal. Conversely, overvaluation paired with bullish sentiment would indicate a sell signal. We take inspiration from the Principle of Contrarian Opinion where we trade in a direction opposite to general sentiment. To evaluate the effectiveness of our strategy, we compare its performance to a simple buy-and-hold strategy using a range of financial metrics that evaluate all aspects of a strategy. The project offers insights into the application of combined valuation and sentiment-based indicators for forecasting stock prices.

2. Strategy and Conceptual Framework

The P/E ratio is a widely used measure of how much the market is willing to pay for \$1 of earnings, which reflects investor sentiment about future earnings growth. By comparing the current P/E ratio with its long-run average, we can determine if a stock is overpriced or underpriced. A lower-than-average P/E ratio could indicate a potential long opportunity, while a higher-than-average P/E ratio could indicate a potential short opportunity.

The SI ratio measures the number of shares that have been shorted in each stock, therefore offering a way to track market sentiment. An increase in short interest indicates an increase in bearish sentiment whereas a decrease in short interest is synonymous with an increase in optimism, or decrease in bearish sentiment, regarding the stock price.

We draw inspiration from the Principle of Contrarian Opinion. Principle of Contrarian Opinion refers to the idea that when speculators in the market are overly optimistic (holding large net long positions), future prices are likely to fall, and when they are overly pessimistic (holding large net short positions), future prices are likely to rise. In their 2015 paper, Kim et al. use the principle to forecast exchange rates by analyzing the positions of speculators, as

recorded in the Commitment of Traders (COT) data. In our strategy we apply this principle to the “sentiment” or short interest data for the Microsoft stock.

Our strategy involves identifying trading signals based on deviations in Microsoft's P/E ratio and SI ratio from their historical moving average. When the P/E ratio drops below a specified threshold (undervaluation) and the SI ratio rises above its moving average (indicating elevated bearish sentiment), we can enter a long position. When the P/E ratio exceeds a specified threshold (overvaluation) and the SI ratio drops below its moving average (reduced bearish sentiment), we can enter a short position. This systematic approach enables us to trade based on fundamental valuation combined with market sentiment, offering a robust framework rather than relying solely on momentum.

Strategy Formulae:

We first define the moving average and the thresholds for overvaluation and undervaluation for the P/E Ratio and SI ratio. Below, n stands for window size and k is the threshold for standard deviation:

- P/E Ratio:
 - Historical Moving Average: $MA\{P/E\} = \frac{1}{n} \sum_{i=1}^n P/E$
 - Undervaluation Thresholds: $MA\{P/E\} - k \times \sigma_{P/E}$
 - Overvaluation Thresholds: $MA\{P/E\} + k \times \sigma_{P/E}$
- SI Ratio:
 - Historical Moving Average: $MA\{SI\} = \frac{1}{n} \sum_{i=1}^n SI$
 - Undervaluation Thresholds: $MA\{SI\} + k \times \sigma_{SI}$
 - Overvaluation Thresholds: $MA\{SI\} - k \times \sigma_{SI}$

Then, we long or short based on trading signals:

- Long Position Signal:
 - When the current P/E ratio is below its undervalued threshold and the current SI ratio is above its undervalued threshold:
 - $P/E \text{ Ratio} < MA\{P/E\} - k \times \sigma_{P/E}$ and
 - $Short \text{ Interest Ratio} > MA\{SI\} + k \times \sigma_{SI}$
- Short Position Signal:
 - When the current P/E ratio is above its overvalued threshold and the current SI ratio is below its overvalued threshold:

$$\blacksquare \quad P/E \text{ Ratio} > MA\{P/E\} + k \times \sigma_{P/E} \text{ and} \\ \text{Short Interest Ratio} < MA\{SI\} - k \times \sigma_{SI}$$

3. Methodology

We sourced our three data series (Microsoft's stock price, P/E ratio, SI ratio) from Bloomberg. Since the SI ratio series has a bi-monthly frequency, we first converted it to a daily frequency using forward-filling. Then, we merged our data based on date and sorted it in an ascending chronological order. Finally, we partitioned the data into a train and test set. The train set starts from 1991-01-15 ending at 2017-12-29, and the test set from 2018-01-02 to 2025-02-18.

For the strategy implementation, we first initialized our model and used moving averages and standard deviations of P/E and SI ratios to establish thresholds for overvaluation and undervaluation. Then we generated buy and sell signals based on these thresholds on our training data, following the contrarian trading approach.

To optimize the hyperparameters of our model, we applied the walk-forward optimization method. We first conducted a hyperparameter search on our training data to identify the optimal combination of hyperparameters for each year. Backtesting was then performed on the test set to simulate how the strategy would have performed in real market conditions.

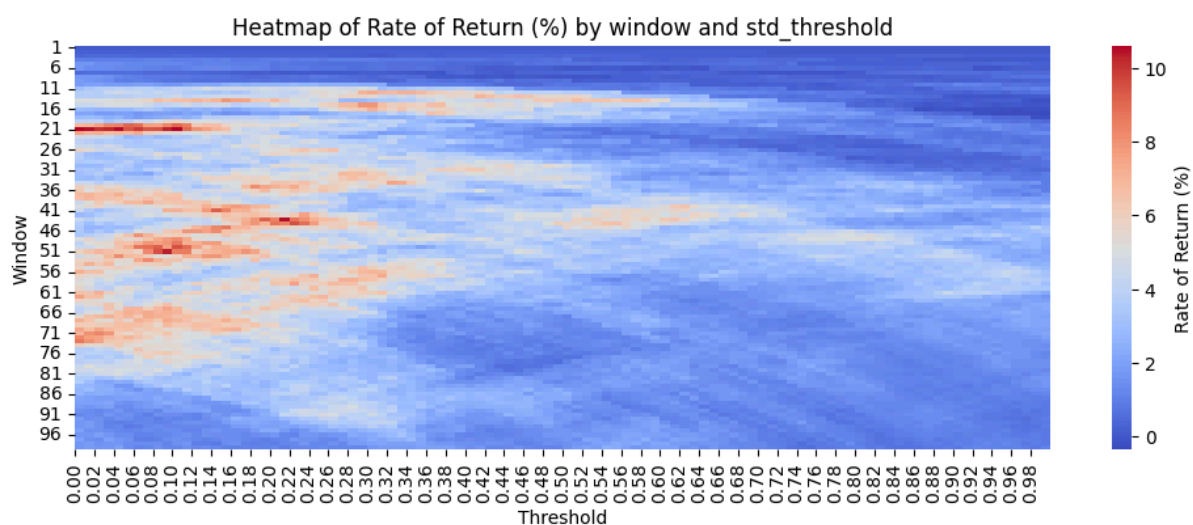
Lastly, we evaluated the strategy performance using a wide range of metrics such as annualized return, alpha, beta, and sortino ratio. For risk free rates in our calculation, we have used the Market Yield on U.S. Treasury Securities at 1-Year Constant Maturity from FRED. Our market benchmark is the S&P500 prices which we have imported from Yahoo Finance. All our metrics are reported on monthly returns to best match the standard of HFRI reports. We also visualized the strategy performance in different aspects to illustrate the strategy more effectively.

4. Results

4.1 Hyperparameters Heat Map on Training Data:

The hyperparameters for the heatmap are window size (n) and standard deviation threshold (k). According to the heatmap, our strategy has the highest return when the window size is

between 20-75 days and the standard deviation threshold is between 0-0.3. And the highest rate of return for our strategy is around 10%.

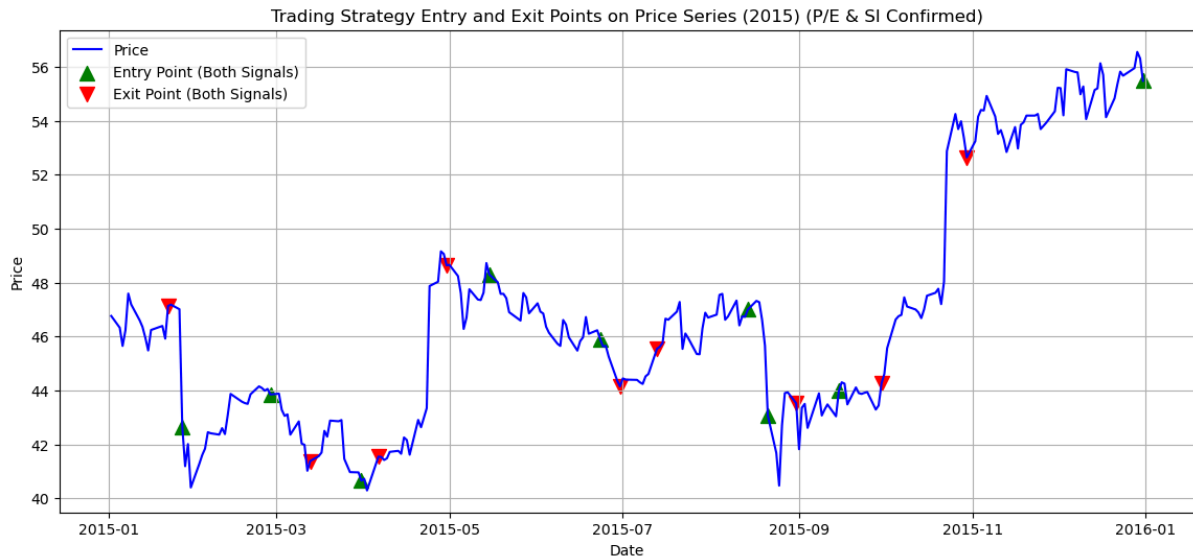


4.2 Visualization of Strategy:

Based on the training set (1991-2017), we zoomed in for a period of time in 2015 to analyze our trading strategy.

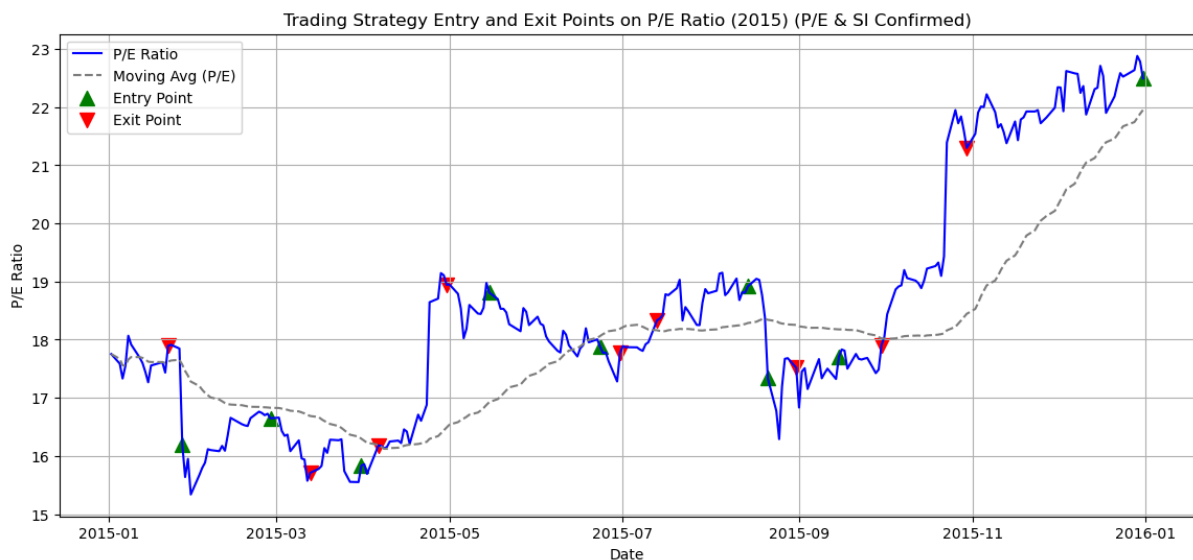
4.2.1 Trading Strategy Entry and Exit Point on Price Series (2015):

This chart illustrates how our trading strategy works in 2015 by marking entry and exit points on the price series. The blue line represents the stock price action throughout 2015. The green upward triangle indicates the entry point where the strategy initiates a long position. This occurs when the P/E ratio falls below the moving average threshold and when the short interest (SI) ratio rises above the moving average threshold. The red downward triangle indicates the exit point, i.e. the position where the strategy closes the position. This occurs when the P/E ratio is above the moving average threshold and the Short Interest Ratio (SI) is below the Moving Average Threshold. In early 2015, the strategy entered long positions following price declines. In mid-2015, prices spiked, triggering exits, and then new positions were entered after prices stabilized. At the end of 2015, prices were experiencing a significant uptrend, with multiple exit points marking the end of the profitable trades.



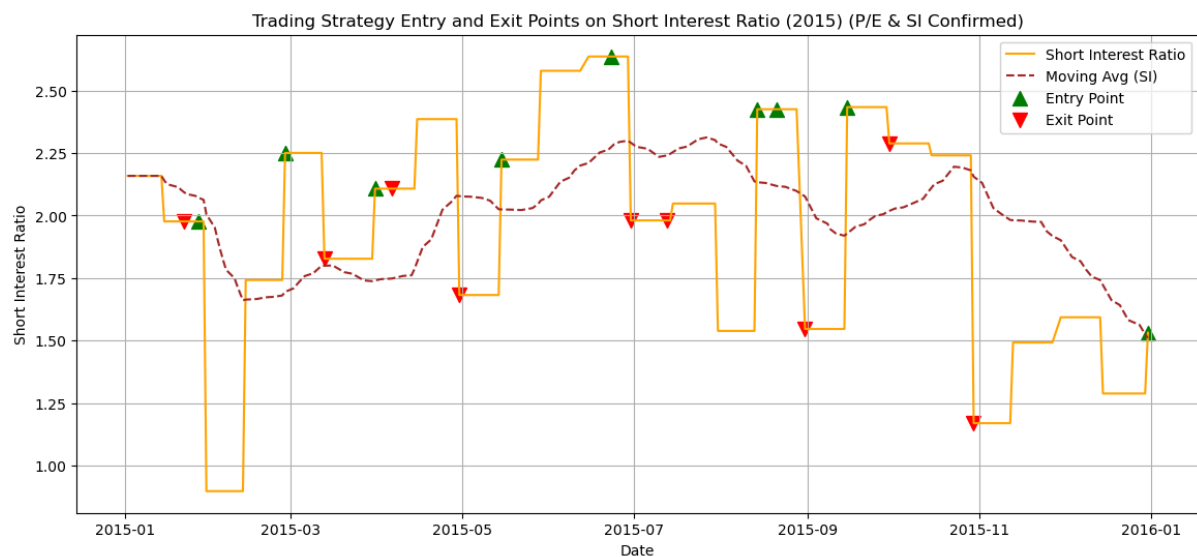
4.2.2 Trading Strategy Entry and Exit Point on P/E Ratio (2015):

The blue line represents the P/E ratio. The gray dotted line represents the moving average of the P/E ratio, which provides a smooth trend and helps avoid small fluctuations. The green triangle (entry point) represents a lower P/E ratio relative to the moving average. Red triangles indicate exit points where the P/E ratio is high relative to the moving average. In this table, entry points tend to occur when the P/E ratio is significantly below the moving average, indicating undervaluation. Exit points occur when the P/E ratio is above the moving average, indicating overvaluation.



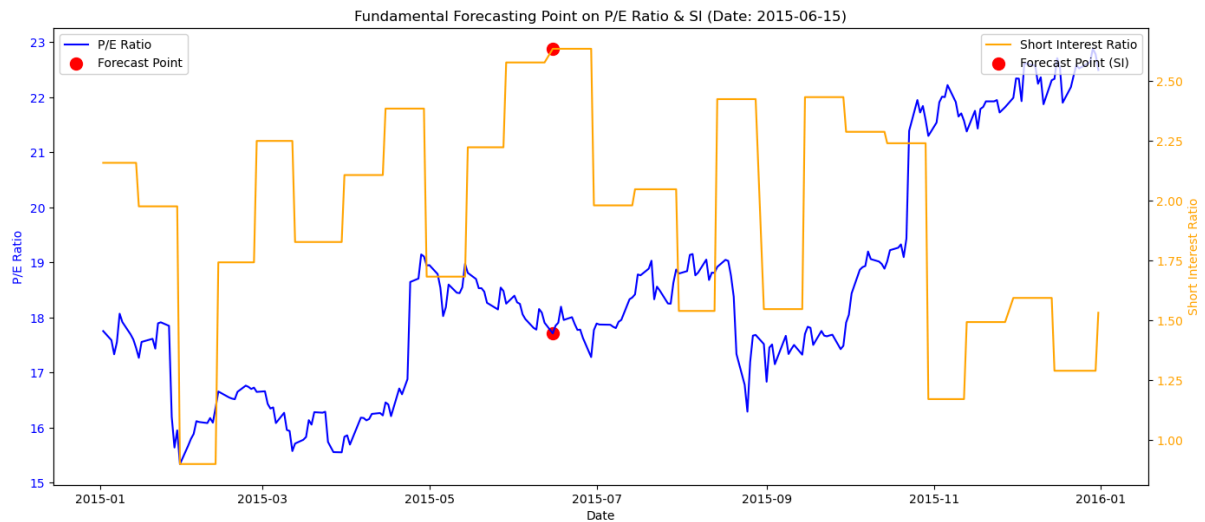
4.2.3 Trading Strategy Entry and Exit Point on Short Interest (2015):

The orange line represents the SI Ratio. The brown dotted line represents the moving average of the SI Ratio. When the green color appears, the SI is higher relative to the moving average, so we can buy; when the red triangle appears, we can sell because at the exit point, the SI is lower relative to the moving average. The key finding of the chart is that a high SI at the entry point indicates that short sellers are overly pessimistic, making the stock attractive to buy. A low SI at the exit point indicates a decline in short-selling interest, making it a good time to sell.



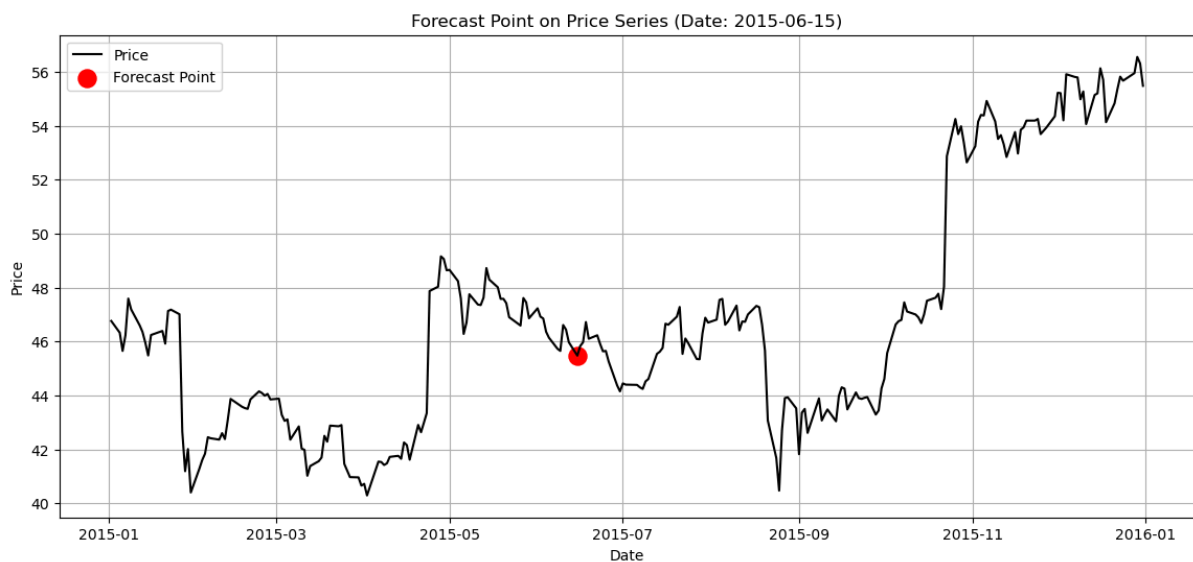
4.2.4 Fundamental Forecasting Point on P/E Ratio With Short Interest (2015-01 to 2015-12):

The chart shows two basic indicators: one is the P/E ratio (blue line) and the other is the Short Interest Ratio (orange line). The red dots indicate that (June 15, 2015) estimation strategies generate signals based on p/e and SI. In the figure, we find that June 15, 2015 is the point at which our model identifies a fundamental shift in stock valuations and sentiment.



4.2.5 Fundamental Forecasting Point on Price Series (2015-01 to 2015-12):

This chart shows the price series for 2015, with the forecast point (marked by the red dot) appearing on June 15, 2015, representing the key moment identified by the guidance strategy. The red dot appears at this point because of changes in the P/E ratio and bearish variance, conditions that indicate that price volatility may occur.

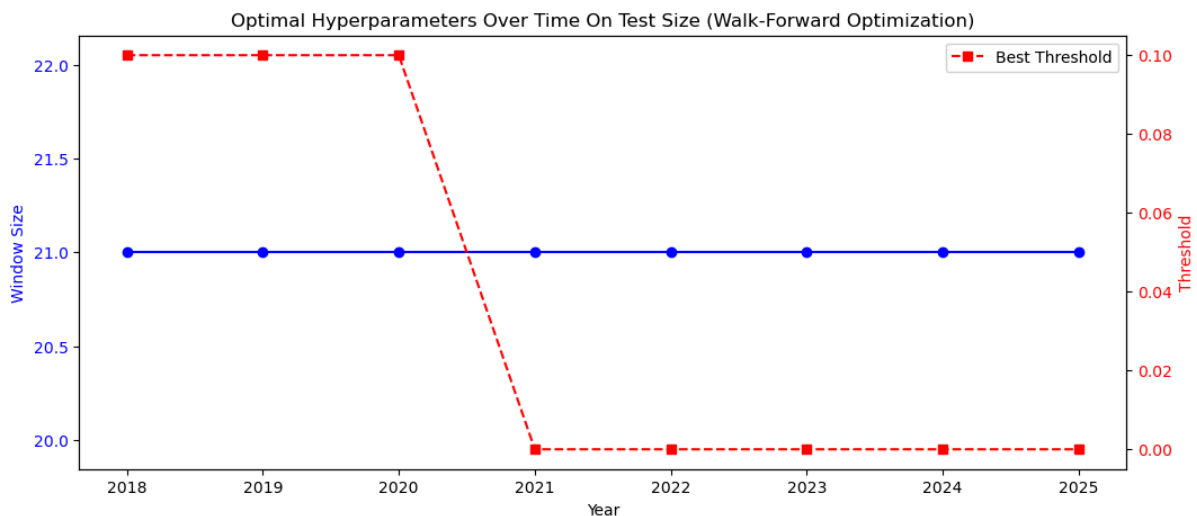


4.3 Back Test with Walk-Forward Optimization:

The chart illustrates the optimal hyperparameters over each year using Walk-Forward Optimization on the test dataset. Among the previous hyperparameters candidates range we generated from a heatmap (windows: 20-75 days, standard deviation threshold: 0-0.3), it

tracks the best hyperparameters: The window size (blue line) remains constant at 21.0 throughout the entire period, showing remarkable stability. Meanwhile, the threshold value (red dashed line) starts at 0.10 from 2018-2020, then drops to 0.00 in 2021 and remains there through 2025. This suggests that while the optimal window size is consistently effective, the model became more sensitive to smaller deviations threshold since 2021, potentially due to the pandemic leading to increased market uncertainty, making a threshold of 0.00 more reasonable in that period.

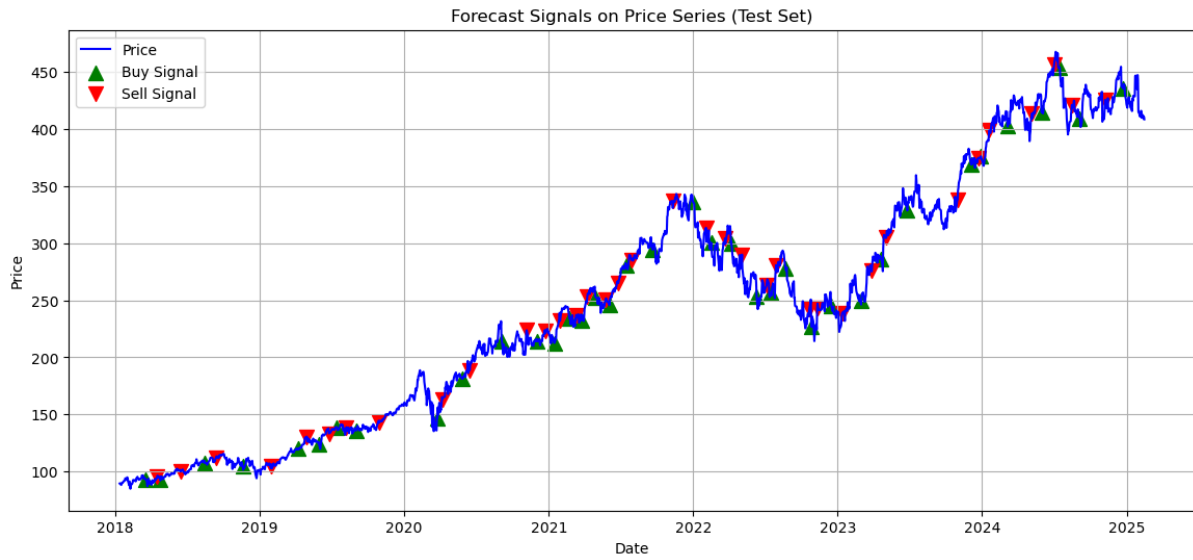
However, in practice, we ultimately choose the hyperparameters 21 and 0.10 to balance stability and robustness. A window size of 21 days provides a stable signal without excessive lag, while a threshold of 0.10 helps reduce noise sensitivity, prevents overfitting to short-term fluctuations, and limits unnecessary trading frequency. Although a lower threshold may have been optimal in highly volatile periods, it could lead to excessive trades and increased transaction costs in more stable market conditions. Therefore, 21 and 0.10 represent a more consistent and risk-controlled choice across different market environments.



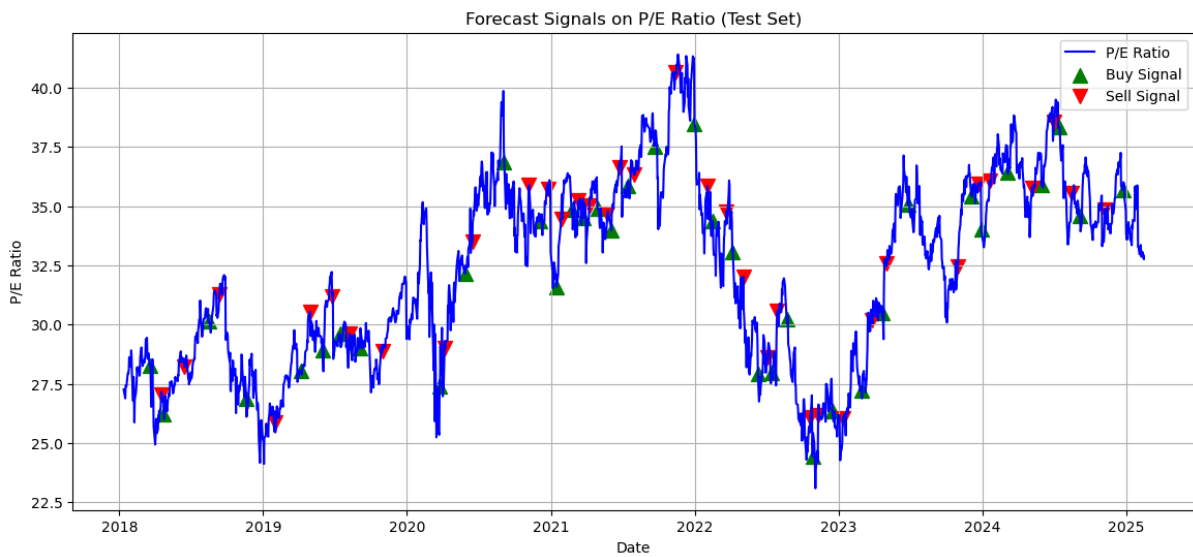
4.4 Forecast Signals over the Back Test Period:

Following are our forecasting signals over the backtesting period (2018-2025). We have followed the strategy algorithm specified in section 2 and have graphed signals on the price series, PE Ratio and Short Interest. In addition to that, from our previous findings, we will use the optimal window size of 21 and the threshold value for 0.1

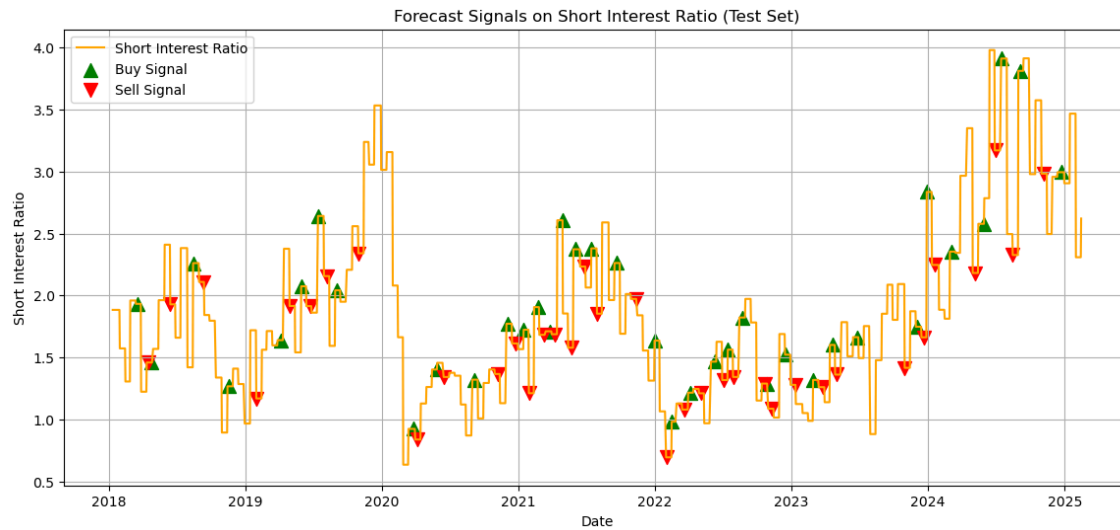
4.4.1 Forecast Signals on Price Series (Test Set):



4.4.2 Forecast Signals on P/E Ratio (Test Set):



4.4.1 Forecast Signals on Short Interest Ratio (Test Set):



4.5 Table of the Performance Metrics:

We will now proceed to analyse our strategy performance. We compare the performance to a buy and hold strategy where we go long on the MSFT stock on the first day (of the backtest) and continue to do so till present day. All our metrics are reported on monthly returns to best match the standard of HFRI reports.

Metric	Forecasting Strategy	Buy and Hold Strategy
Annualized Return	0.123073	0.208791
Annualized Std Dev	0.232232	0.206403
Alpha	0.009798	0.009715
Beta	0.135668	0.911798
Up Alpha	0.004729	0.017725
Up Beta	0.304603	0.733298
Down Alpha	-0.011055	0.009796
Down Beta	-0.278484	0.960792
Max Drawdown	0.002021	0.003657
Drawdown Date	2020-10-31	2022-10-31
Highest Monthly ROR	0.183693	0.176291
Lowest Monthly ROR	-0.109267	-0.109267
Gini Coefficient	0.670901	0.658951
Return on Account	499.52385	277.555469
Skewness	0.401569	0.070135
Sortino Ratio	0.288502	0.535842
Sharpe Ratio	0.162928	0.292234

Table 1: Performance Metrics Comparison

Comparing the two strategies we notice that:

1. Buy and Hold Outperforms in Returns: The Buy and Hold Strategy generates higher Annualized Returns (20.88%) compared to our Forecasting Strategy (12.31%), making it more attractive for long-term investors.
2. Forecasting Strategy is Less Correlated: Our Forecasting Strategy has a much lower Beta (0.1357), indicating it is less sensitive to market movements. This could be advantageous in volatile or bear markets.
3. Buy and Hold Manages Downside Risk Better: The Buy and Hold Strategy has a higher Sortino Ratio (0.5358) and better performance during downturns (Down Alpha and Down Beta), making it more resilient during market declines.
4. Forecasting Strategy Has Higher Skewness: Our Forecasting Strategy has a higher Skewness (0.4016), indicating a higher likelihood of extreme positive returns, which could appeal to risk-seeking investors.

5. Forecasting Strategy Has Lower Max Drawdown: Our Forecasting Strategy has a lower Max Drawdown (0.0020) compared to the Buy and Hold Strategy (0.0037), indicating better performance during the worst periods.
6. Forecasting Strategy Has Higher Return on Account: Our Forecasting Strategy has a higher Return on Account (499.52) compared to the Buy and Hold Strategy (277.56), indicating better risk-adjusted profitability.
7. Forecasting Strategy Has Slightly Higher Gini Coefficient: Our Forecasting Strategy has a slightly higher Gini Coefficient (0.6709) compared to the Buy and Hold Strategy (0.6589), indicating slightly more inequality in the distribution of returns.
8. Buy and Hold Performs Better in Upswings: The Buy and Hold Strategy has higher Up Alpha (0.0177) and Up Beta (0.7333), indicating better performance during market upswings compared to our Forecasting Strategy (0.0047 and 0.3046).
9. Buy and Hold is Less Volatile: The Buy and Hold Strategy has a lower Annualized Standard Deviation (0.2064) compared to our Forecasting Strategy (0.2322), despite its higher sensitivity to market movements (Beta).
10. Forecasting Strategy Offers Lower Market Correlation: The low Beta (0.14) of our Forecasting Strategy makes it less dependent on market trends, which could be beneficial in uncertain or declining markets.
11. Buy and Hold is More Consistent: The Buy and Hold Strategy has a lower Skewness (0.07), indicating more consistent returns compared to our Forecasting Strategy (0.40), which has a higher likelihood of extreme positive returns.
12. Forecasting Strategy Offers Low Risk Adjusted Return: Our strategy (0.16) has a lower Sharpe Ratio as compared to the Buy & Hold (0.29) which gives a higher return for each unit of risk taken

Lastly, our results from the Weighted Directional Test and Binomial Forecast Accuracy Test indicate that our forecasting strategy's performance is not statistically significant. We achieved a Weighted Directional Accuracy (WDA) of 0.52 and an accuracy of 0.51, both of which are close to the 50% expected for random guessing. The p-values of 0.3314 (Weighted Directional Test) and 0.1716 (Binomial Test) suggest that our strategy's predictions are not significantly better than random.

5. Conclusions

Compared to the S&P 500, our strategy exhibits low correlation with the market and low excess returns. Even when we segment market returns into positive returns to calculate up alpha and up beta, respectively, we observe the same trend. However, when the market sees a downturn, our strategy has a negative beta indicating that it moves in a direction opposite to the market. This is different for our Buy & Hold down beta which is strongly correlated to the market in times of a downturn. While it does not perform well as a standalone strategy, our strategy, with some tweaks, could be used in a portfolio to hedge against market downturns. It is also important to note that our hyper-parameters were less than optimal for the years 2021-2025, a revision for them could be considered to further refine the strategy.

For future improvements, additional market variables could be included to enhance predictive accuracy. For example, we could use social media sentiment or news sentiment as our alternative sentiment indicator. To mitigate downside risks, we could also add stop-loss techniques into our strategy.